

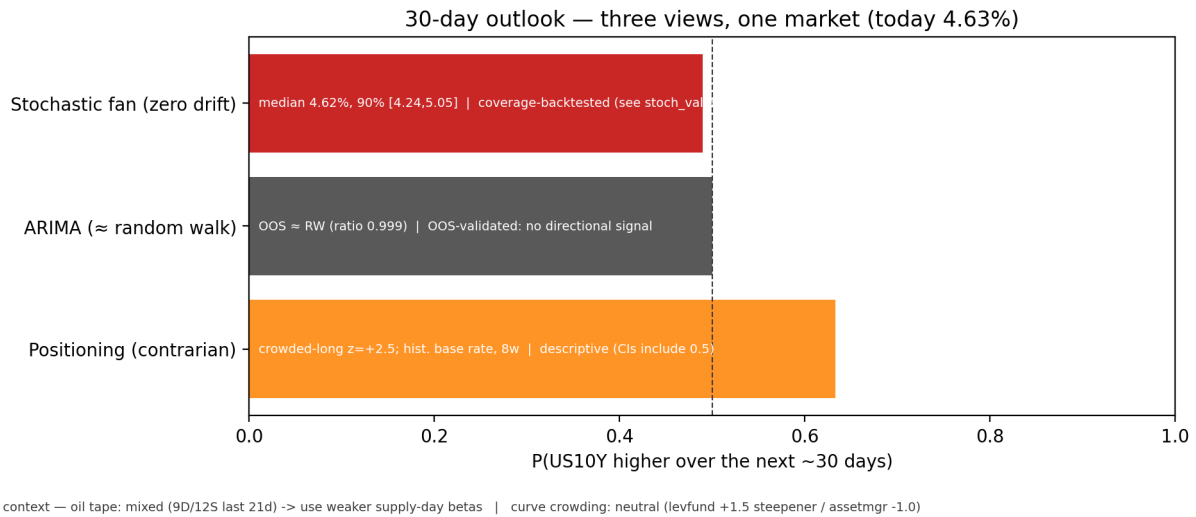
US Rates Research Pack

Consolidated, audited session output: CFTC positioning & curve crowding · positioning→yields evidence · US10Y forecasting (ARIMA, GBM/Heston, walk-forward validation) · oil→rates conditional sensitivity · audit of revisions. All data free/public (CFTC, FRED, NY Fed); every claim carries an evidence grade: [V] = out-of-sample validated, [S] = statistically significant (HAC), [D] = descriptive only.

1. Executive summary — verified headlines

Claim	Value	Grade
Positioning: crowded net-LONG duration extreme	composite $z = +2.49$, 89.7th pctile (2026-07-14)	[D]
driver: asset managers long the long end; dealers mirror-short	AM $z +1.21$; lev funds -0.25	[D]
Crowded-long extremes preceded higher yields. . .	P(up) 0.53/0.63/0.66/0.74 at 4/8/13/26w	[D]
. . . but NOT statistically robust	all 95% CIs include 0.5; Granger $p \geq 0.80$	[S-tested, fails]
US10Y point forecasting: nothing beats a random walk	ARIMA/RW RMSE 0.9985, DM $p=0.62$	[V]
30-day fan (validated widths, zero drift)	median 4.62%, 90% [4.24, 5.05], P(up) 49%	[V (coverage)]
trend drift is WORSE than zero drift OOS	RMSE 0.316 vs 0.287, DM $p=0.07$	[V]
all fan conditioning rejected OOS at 30d	CRPS 16.05bp base vs 16.05–16.20 variants	[V]
Desk: barrier / tail numbers (10k paths)	P(touch 5.00%)=11%, P(touch 4.25%)=9%; 10M 10y VaR95 –330k	[V (widths)]
Oil moves rates only via breakevens, and barely	$\beta_{10y} = +0.26\text{bp}/1\%$ ($t=6.6$, $R^2=1.4\%$); real leg $t=0.4$	[S]
big-oil days: real & breakeven legs offset	opposite signs 60% of days, real larger	[D]
sensitivity saturates with oil vol	interaction $t = -5.0$; oil > 5% days: mean $\Delta y = -0.7\text{bp}$	[S]
demand-day beta significant, supply-day not	+0.215 ($t=2.6$) vs +0.132 ($t=1.6$), 2016+	[S / fails]
Curve books are FLAT in raw DV01 (both categories)	lev -0.1 , AM -5.0 \$mm/bp front–long	[D]
the +1.46 “steepener” $z =$ unwind momentum, not a book	raw flat; z vs 3y history	[D, corrected]

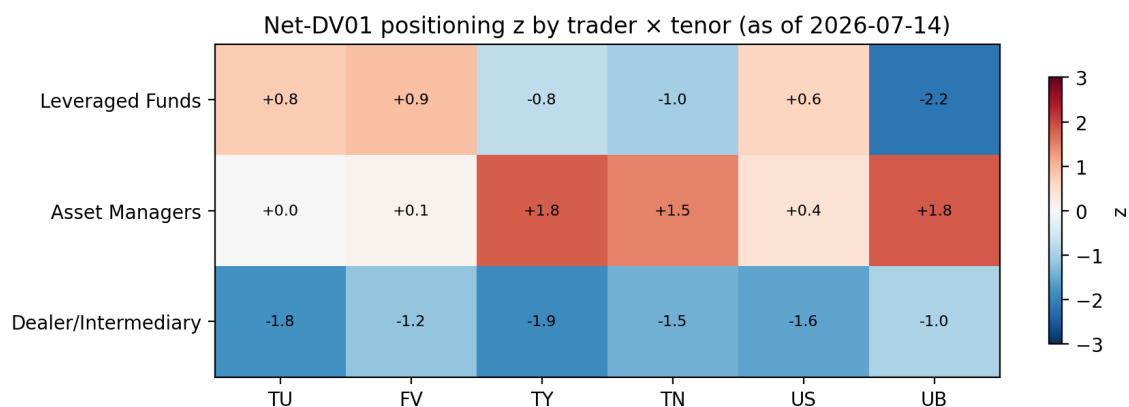
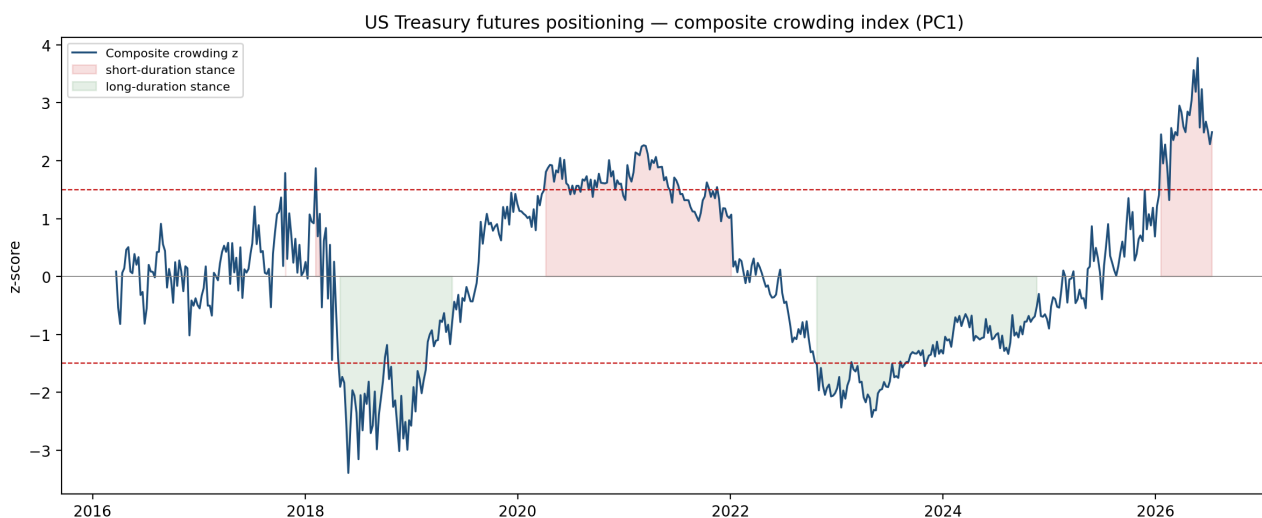
One-line outlook. No honest directional edge on the 10Y over 30 days (P(up)≈49–50% from every validated model); the crowding backdrop leans selloff [D]; the validated uncertainty band is 4.24–5.05%; oil is a second-order input on a mixed/supply tape; curve positioning is momentum-steepening but books are flat.



2. Positioning: crowding score and curve read

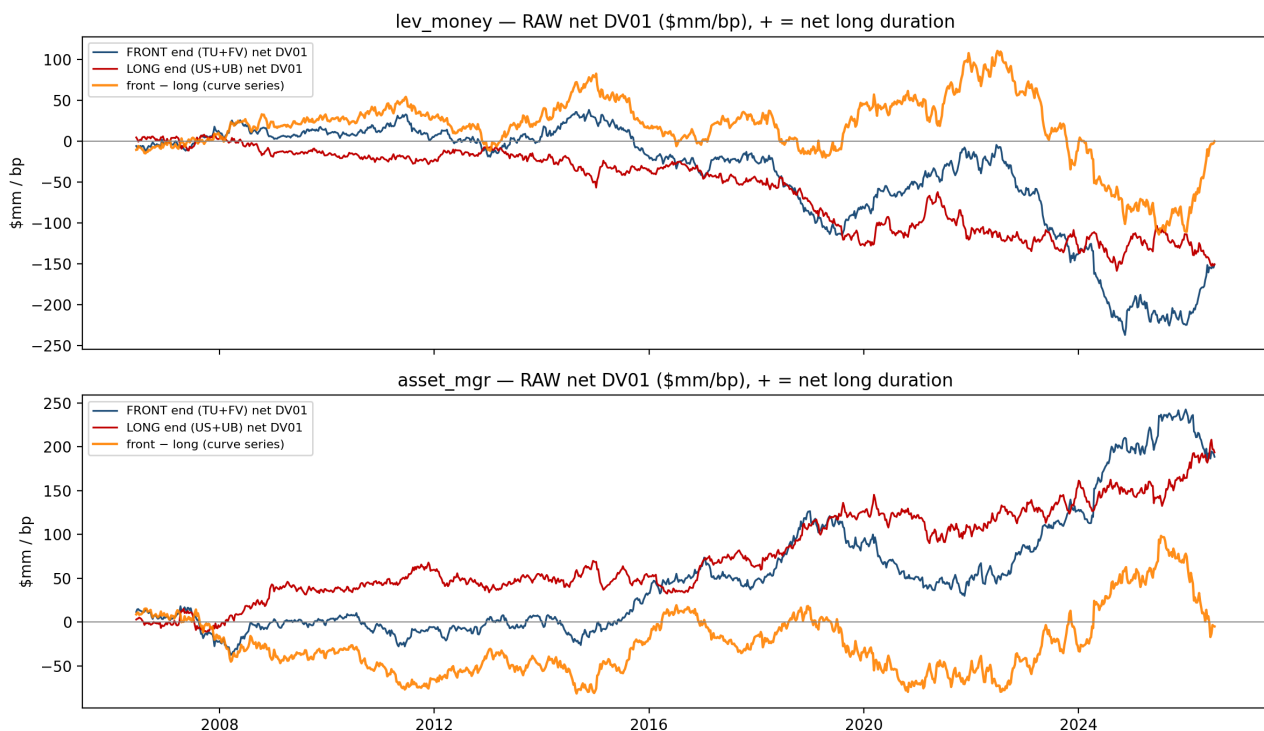
Method. CFTC TFF net positions (TU/FV/TY/TN/US/UB) $\rightarrow$ DV01 $\rightarrow$ per-category z-scores $\rightarrow$ PCA composite (crowding z ; + = crowded net long duration). Causal publication-lagged timeline throughout; predictive overlay embargo-purged (leakage-free walk-forward AUC 0.554 — weak, treated as context only).

Current read (2026-07-14). Composite $z = +2.49$ (89.7th pctile), a multi-year crowded-long extreme; asset managers carry it (long TY/TN/UB), dealers mirror-short, lev funds $\approx$ neutral outright. Stance: contrarian short-duration [D].



Curve positioning — the corrected two-dimension read. The per-category curve z (front TU+FV minus long

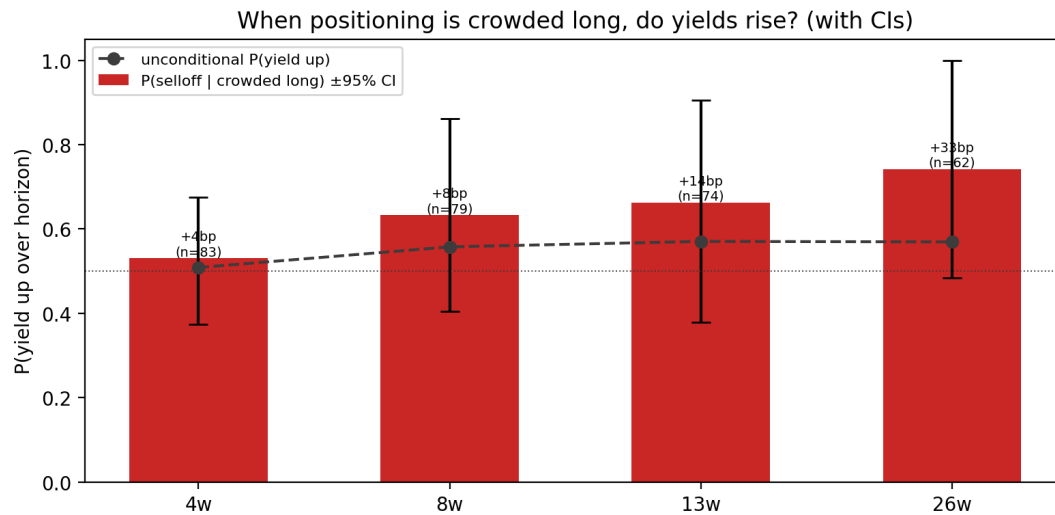
US+UB net DV01, vs 3y history) shows lev funds at +1.46; but the RAW books are flat: lev funds front -150.9 vs long -150.8 \$mm/bp (difference -0.1); asset managers $+188.4$ vs $+193.5$ (-5.0). The z therefore measures the sharpest steepener-DIRECTION unwind in 3 years (from a large 2022–24 flattener book back to flat), not a crowded steepener stock. Historically, extremes of this z preceded curve FLATTENING ($P(\text{flatten}|\text{steepener-}z > 1.5) = 0.74/0.80/0.78$ at 4/8/13w, mean $-7/-13/-20$ bp; flattener- z extremes showed continuation, $0.57\text{--}0.65$) — a positioning-momentum-exhaustion pattern [D: full-sample, clustered episodes].



3. Does crowded positioning predict yields? (the honest evidence)

Point estimates lean contrarian and strengthen with horizon (table), but the effect is NOT statistically robust: block-bootstrap CIs include 0.5 at every horizon, Granger causality is absent ($p = 0.80\text{--}0.95$), Newey–West forward regressions are weak ($t = 0.3\text{--}1.1$, $R^2 \leq 3.5\%$), and the crowded weeks form few independent episodes (the current one is 21 consecutive weeks). Sub-period analysis concentrates the effect in 2020–22 and 2023–26.

Horizon	n	P(yield up)	95% CI	mean Δy	unconditional
4w	83	0.53	[0.37, 0.67]	+3.7bp	0.51
8w	79	0.63	[0.41, 0.86]	+7.9bp	0.56
13w	74	0.66	[0.38, 0.91]	+14.1bp	0.57
26w	62	0.74	[0.48, 1.00]	+32.9bp	0.57

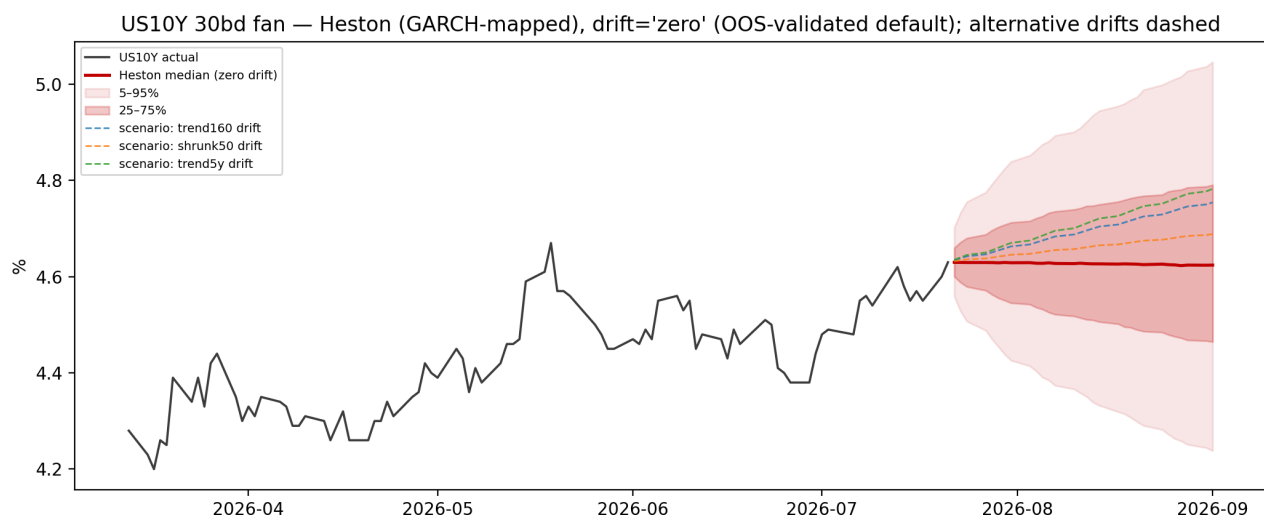


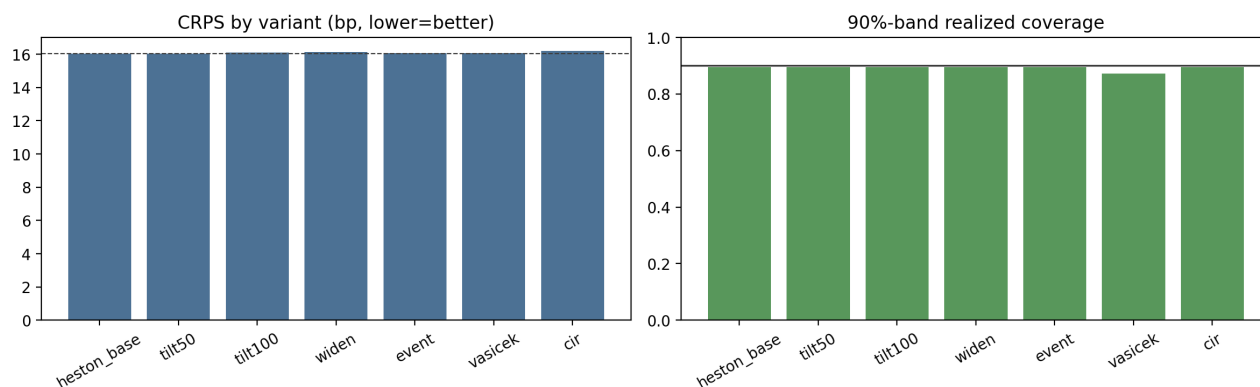
4. US10Y forecasting: ARIMA, stochastic fan, and the validation program

ARIMA [v]. DGS10 is $I(1)$; recursive walk-forward ARIMA(0,1,2) one-step OOS RMSE ratio vs random walk = 0.9985 (DM $p = 0.62$): indistinguishable from RW. Adding the positioning composite as ARIMAX exogenous input does not help (ratio 0.9993, DM $p = 0.80$).

Stochastic fan [v for widths]. The yield is modelled as an asset price (GBM / Heston SV, 10,000 antithetic paths, 30bd). Heston parameters are mapped from a GARCH(1,1) MLE (v_0 = the filter's current conditional variance: 15.1%/y ann. vol; long-run 16.9%/y; $\kappa \approx 0.5/y$; $\rho = -0.08$). Walk-forward validation at 125 non-overlapping origins (2011–2026):

- **Coverage is honest:** the 90% band covered 89.6% of realized 30bd outcomes (Kupiec $p = 0.88$); PIT uniform (KS $p = 0.65$).
- **Drift: zero (RW) wins;** the since-Dec-2025 trend drift is the WORST candidate (RMSE 0.316 vs 0.287; DM $p = 0.072$; its P(up) Brier 0.295 is worse than know-nothing 0.25). The earlier trend-drift headline “P(up)=70%” is retracted; the validated number is 49%.
- **All conditioning rejected at 30d** (CRPS scoreboard): positioning tilt (16.05bp, no gain; active 18/125 origins), stress widening (16.12, worse — GARCH v_0 already carries the vol state), calendar/event vol (16.06, diluted at ~ 3 event days of 30; expected to matter at 1w), Vasicek (equal CRPS but under-covers 0.87), CIR (16.20). The base fan is a proven local optimum at this horizon.



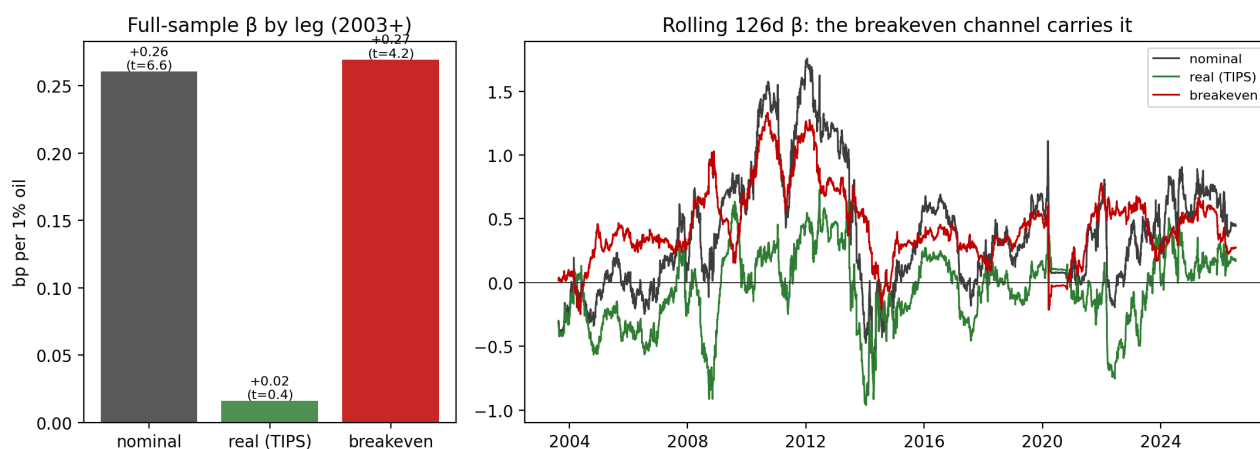


Desk numbers (from the validated fan). Day-30 median 4.62% (± 0.2 bp MC s.e.), 90% band [4.24, 5.05]%; $P(\text{touch } 5.00\%) = 10.8\%$, $P(\text{touch } 4.25\%) = 8.6\%$; long 10y 10M notional 30d VaR95 ≈ -330 k (DV01 ~ 793 /bp per 1M). Mean-reversion benchmark: Vasicek half-life 182d toward $\theta = 4.35\%$ (under-covers OOS; kept as comparator, not default).

5. Oil and rates: a conditional, channel-specific relationship

The puzzle quantified [S]. Daily $\beta(\Delta y_{10} \sim \text{oil}) = +0.26$ bp per 1% ($t = 6.6$) but $R^2 = 1.4\%$: a 5% oil move maps to ~ 1.3 bp. Monthly $+0.36$ ($t = 2.6$); Brent similar.

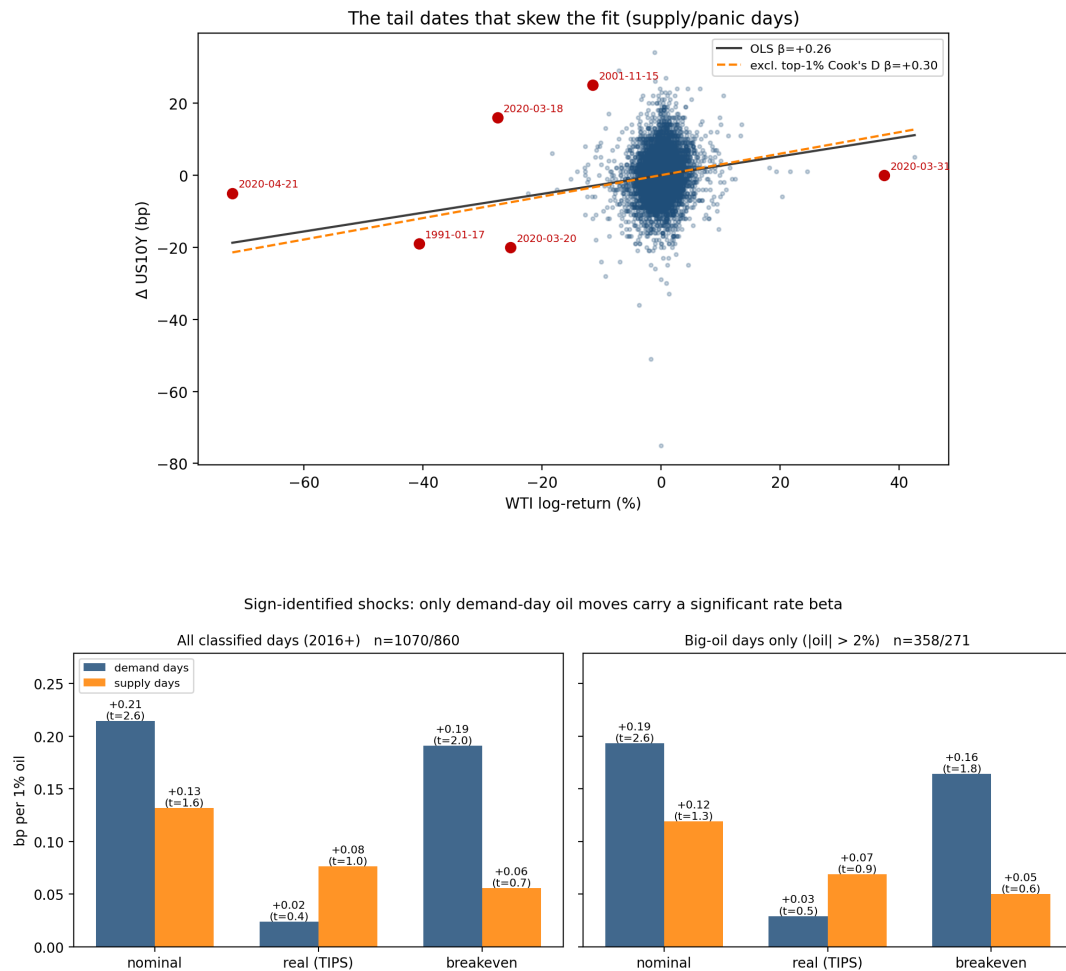
Mechanism: offsetting channels [S]. The entire nominal effect runs through breakevens ($\beta_{BEI} = +0.27$, $t = 4.2$, $R^2 = 5.3\%$); the real leg is nil on average ($t = 0.4$). Granger: oil $\rightarrow$ BEI $p = 0.003$; oil $\rightarrow$ real $p = 0.44$; strong reverse feedback rates $\rightarrow$ oil ($p \approx 0$). On $|\text{oil}| > 3\%$ days ($n = 737$) the two legs move in OPPOSITE directions 60% of the time (corr -0.32) with the real leg larger (4.7 vs 3.5bp): the growth channel cancels the inflation channel exactly when oil moves most.



Regimes and saturation [S]. A 2-state Markov switch puts the LOW-beta state ($\beta = 0.21$, 37% of days, $\sigma = 8$ bp/d) in high-stress conditions (VIX 23 vs 17; elevated oil and rate vol; fires in 2008–09, 2011–12, 2020, 2022–23); the calm state carries $\beta = 0.33$. The only significant interaction is oil volatility ($t = -5.0$): per-unit sensitivity roughly halves as oil vol doubles, so bp-response is nearly regime-invariant (~ 0.55 – 0.68 bp/day) — the response SATURATES.

Shock type (2016–2026, FRED SP500 classifier) [S/tails]. Demand-day $\beta = +0.215$ ($t = 2.64$, significant, via breakevens); supply-day $\beta = +0.132$ ($t = 1.61$, indistinguishable from zero); ordering as hypothesized but the magnitude gap is modest — the sharp distinction is significance. Current 21d tape: 9 demand / 12 supply = mixed $\Rightarrow$ apply the weak supply-day betas.

Datapoints that skew the fit [D]. Cook's-distance leaders are supply/panic days — 2020-04-21 (oil -72% , rates -5 bp), 2020-03-18 (-27.5% , $+16$ bp wrong-signed), 1991-01-17 (Gulf War), Nov-2008 — and they DRAG the unconditional beta DOWN: OLS $+0.261 \rightarrow +0.298$ excluding the top-1% Cook's D (Huber $+0.267$). Conditional on $|\text{oil}| > 5\%$ ($n = 349$): mean $\Delta y_{10} = -0.7$ bp ≈ 0 . (The 2020-04-20 negative-WTI print is excluded from returns by construction.)



6. Audit: inconsistencies found and resolutions

- P(up) 70% → 49% (retraction).** The first fan used the since-Dec-2025 trend drift; walk-forward validation showed that drift is WORSE than zero OOS (DM $p = 0.07$). The validated headline is P(up)=49%.
- “Crowded steepener” → momentum, flat book (correction).** The +1.46 curve z is relative to a 3y window dominated by a flattener book; raw front–long DV01 is -0.1 (lev) / -5.0 (AM) \$mm/bp. Labels now separate book tilt (flat) from positioning momentum (steepener-direction). The historical “steepener→flatten 74–80%” result is correspondingly a momentum-exhaustion statement.
- Heston calibration v1 → v2.** The first calibration used a smoothed rolling-variance proxy ($\theta = 28\%/y$, $\kappa = 3.2/y$, $\rho = -0.03$ attenuated); v2 maps a GARCH(1,1) MLE ($\theta = 16.9\%/y$, $\kappa = 0.5/y$, $\rho = -0.08$). v2 is the reported model; its coverage was validated directly.
- Bootstrap CIs vs point estimates (flag).** All three v2 variance-dynamics point estimates sit OUTSIDE their block-bootstrap 95% CIs (κ 0.5/y vs [1.5, 10.6]/y; θ 16.9% vs [26, 54]%; ξ 0.00037 vs [0.0018, 0.0075]): the 63d block bootstrap breaks GARCH long-memory and biases persistence down. Read the CIs as uncertainty MAGNITUDE, not centered intervals; parameter uncertainty is large (Feller CI includes violation).
- Vintage drift in quoted numbers.** Composite z +2.51/91st pctile was the 2026-06-30 vintage; the current CFTC vintage (2026-07-14) gives +2.49/89.7th. Data vintages differ by design: CFTC 07-14, FRED yields 07-21/22, oil 07-20.
- Positioning predictive AUC 0.5585 → 0.5543.** The small drop is the overlapping-label embargo purge (leakage fix), not a model change.
- Rounding.** “Coverage 90%” is 89.6% (Kupiec $p = 0.88$); validation base-rate-up is 0.496 (reported earlier as “0.50”).
- Cross-checks that PASSED.** Tool vs study conditional base rates agree to 3dp (0.530/0.633/0.662); oil unconditional, decomposition, Markov, interaction, influence and shock-type numbers all match their JSONs; ARIMA and bridge ratios match; the outlook panel numbers match the underlying JSONs.

9. **Framing note.** The oil regime results (Markov low-beta in stress; supply-day beta ≈ 0 ; vol saturation) are three related but DISTINCT identifications that point the same way; they are complementary evidence, not one test.

7. Data, methods, reproducibility

Sources: CFTC TFF (gpe5-46if, keyless), FRED (DGS2/5/10/30, DFII10, T10YIE, T5YIFR, DCOILWTICO, DCOIL-BRENTEU, VIXCLS, DTWEXBGS, DFF, SP500), NY Fed FR2004. Methods: DV01-normalized positioning with causal z-scores and publication lags; HAC/Newey–West inference; block-bootstrap CIs; Markov-switching ML; Cook’s-distance influence; expanding-window walk-forward with horizon embargo; CRPS/log-score/Kupiec/PIT for distributional validation. Reproduce (repo root, PYTHONPATH=.): `python -m positioning.run_positioning` · `python -m rates_study.run --report` · `python -m rates_study.stoch_validation` · `python -m rates_study.stochastic` · `python -m oil_rates.run --report`. **Operational warning:** the entire tree (all studies + the VBA work below) is uncommitted in git.

Appendix: VBA PnL Explainer workstream (status)

Separate from the studies, this session also fixed the Excel PnL Explainer: (i) **critical** — `WrapIfPresent` was an infinite self-recursion (62 call sites; the true source of the “Out of stack space” failures); fixed in `modPNL_reorganized.txt`. (ii) `EnableEvents` state-restore normalized in steps 3B/3D/3E/3F. (iii) Bonds Benchmark/DCC blanking fixed (freeze removed; day-count synonyms). (iv) `PayFixed` populated via `NETPAYIND`→Y/N keyed on `LinkedISIN`. (v) Swap Bloomberg DV01 in col BJ (ABS×sign convention) repointed into PnL attribution. `modPNL_finalized.txt` carries these plus dead-code removal; the larger refactor (bulk-array writes, module split) is paused with state saved. Run-time items still requiring a live Bloomberg terminal: `NETPAYIND` encoding, swap DV01 field (`dv01` vs `SW_EQV_BPV`), benchmark BDP mnemonic.

Generated from the session’s audited result files. Evidence grades: V = out-of-sample validated; S = statistically significant (HAC); D = descriptive. Every revised claim appears in Section 6 as a revision, not silently replaced.